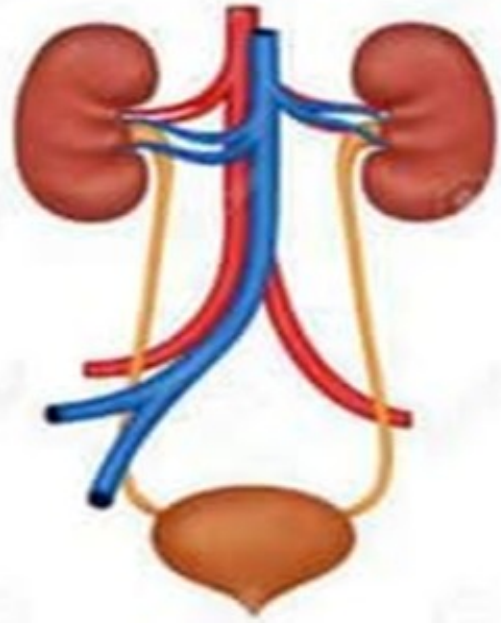
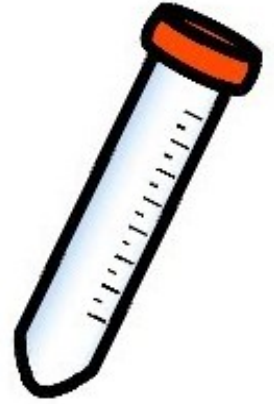




RENAL FUNCTION TEST



Objective

- ❖ **Define renal function test**
- ❖ **Identify the biochemical kidney function tests**
- ❖ **Interpret the kidney function tests properly**

Introduction

- ❖ **Renal function tests (RFT)** are urine or blood tests that evaluate how well our kidneys are working.
- ❖ Most of these tests measure GFR which assesses how efficiently your kidneys clear waste from your system.
- ❖ Used to:
 - Assess Kidney Health
 - Detect Kidney Disorders
 - Monitor Disease Progression
 - Guide Treatment Decisions

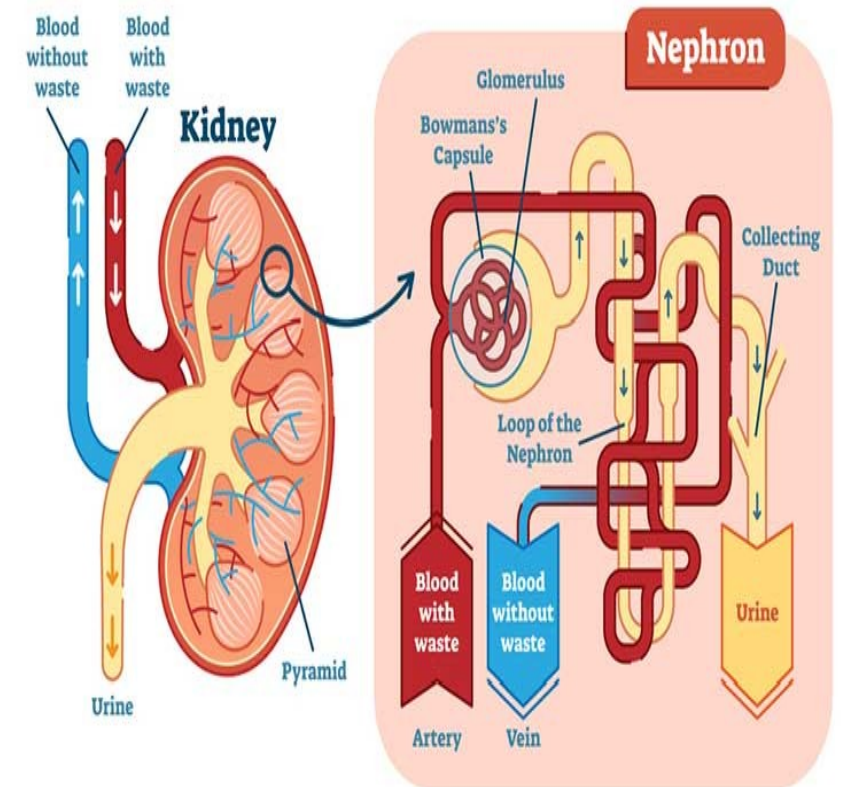
Function of the kidney

➤ Regulation of :

- water and electrolyte balance.
- acid base balance.
- arterial blood pressure.

➤ **Excretion** of metabolic waste products and foreign chemicals.

➤ **Hormonal Function:** Secretion of erythropoietin & activation of vitamin D and activation of angiotensinogen by renin



Types of Renal function test

❖ There are a number of functional test of the kidney

- **Blood Tests**

- Serum Creatinine
- Blood Urea Nitrogen (BUN)
- Estimated Glomerular Filtration Rate (eGFR)

- **Urine Tests**

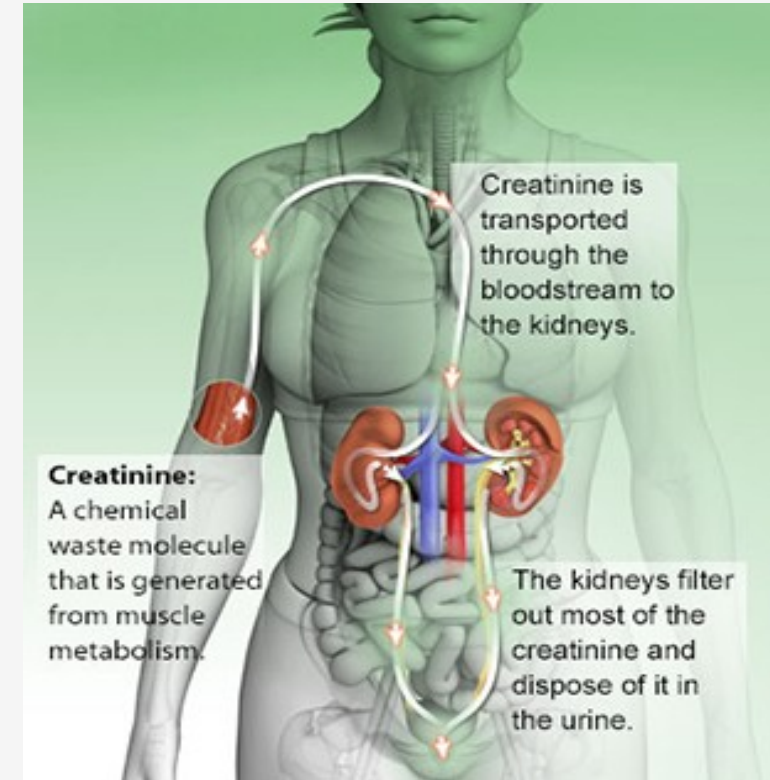
- Urinalysis
- 24-hour Urine Collection
- Microalbuminuria

- **Imaging Studies**

- Ultrasound
- CT Scans
- MRI

Serum Creatinine

- ❖ Creatinine is produced from the breakdown of **creatine** and **phosphocreatine**, which are primarily found in muscle tissue.
- ❖ Is an indicator of changes in GFR.
- ❖ Serum creatinine levels can be lower in conditions where muscle mass is reduced.
- ❖  GFR results in an in the  SCr concentration.



Cont'd

- ❖ Serum creatinine are affected partly by:
 1. **The amount of muscle tissue** : Men >> women.
 2. **Protein in diet** : Vegetarians tend to have lower creatinine levels in blood.

- ❖ This test is a more accurate and sensitive marker for kidney disease than blood urea nitrogen (BUN).

- ❖ In case of chronic kidney disease the **BUN-to-creatinine ratio** provides valuable additional insights into kidney function.

Normal SCr Range

Group	Serum Creatinine Range	BUN-to-Creatinine(B/C) Ratio
Adult Men	0.7 - 1.3 mg/dL (62 - 115 µmol/L)	10:1 to 20:1
Adult Women	0.6 - 1.1 mg/dL (53 - 97 µmol/L)	10:1 to 20:1
Children	0.3 - 0.7 mg/dL (26 - 62 µmol/L)	10:1 to 20:1
Newborns	0.2 - 0.4 mg/dL (18 - 35 µmol/L)	10:1 to 20:1

- Clinical Significance of SCr are:

❖ **Kidney Function Assessment** : Elevated SCr indicates ↓ kidney filtration/dysfunction.

❖ **GFR Estimation** : to calculate glomerular filtration rate.

❖ **Monitoring Renal Health** : Tracks progression of chronic kidney disease (CKD) or acute kidney injury (AKI).

Factors influencing SCr values

1

Increased SCr

- Kidney dysfunction
- Rhabdomyolysis
- High muscle mass
- Congestive heart failure
- Acromegally

2

Decreased SCr

- Low muscle mass
- Malnutrition
- Aging
- Pregnancy
- Hyperthyroidism

Blood Urea Nitrogen(BUN)

- ❖ **Urea nitrogen** is an end product of protein and amino-acid metabolism.
- ❖ It is produced solely by the liver, **is transported in the blood, and is excreted by the kidneys.**
 - Decrease in BUNinability of liver to synthesize Urea
 - Increase in BUN..... inability of kidney to filter out
- ❖ **Acute or chronic renal failure** is the most common cause of an elevated BUN.

Normal BUN Range

Group	Normal BUN Range
Adult (both men and women)	6 - 24 mg/dL
Children	5 - 18 mg/dL
Newborns	2 - 10 mg/dL

❖ **Several nonrenal factors** such as

- unusually high protein intake and
- conditions that increase protein catabolism (or upper GI bleeding)

can increase the BUN concentration.

Factors influencing BUN values

1

Increased BUN

- Acute and Chronic renal failure
- Congestive heart failure
- Diabetes
- Hypovolemia
- Excessive protein ingestion

2

Decreased BUN

- Inadequate dietary protein
- Low-protein/high-carbohydrate diet
- Malabsorption syndromes
- Pregnancy

Estimated Glomerular Filtration Rate (eGFR)

- ❖ Estimated Glomerular Filtration Rate (eGFR) is a calculation used to estimate how well the kidneys are filtering blood
- ❖ It measures the rate at which the kidneys clear creatinine, a waste product, from the blood.
- ❖ Used to diagnose and stage chronic kidney disease (CKD).
- ❖ **Importance/Clinical Significance:**
 - Assess and monitor Kidney Function
 - Staging Chronic Kidney Disease (CKD)

Cont'd

- ❖ Typical eGFR is **90 mL/min/1.73 m²** or higher, indicating normal kidney function.
 - eGFR < **90 mL/min/1.73 m²** = filtering of waste from the blood is at a slower rate
 - eGFR > **120–130 mL/min/1.73 m²** : kidneys are working harder than normal
- ❖ The most commonly used and convenient formula used for determining eGFR is **Cockcroft-Gault Equation**

$$CrCl \left(\frac{ml}{min} \right) = (140 - age) \frac{BW}{Scr * 72} \times (0.85 \text{ if women})$$

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- ❖ The **KDIGO** (Kidney Disease: Improving Global Outcomes) CKD classification system stages kidney function based on eGFR values:

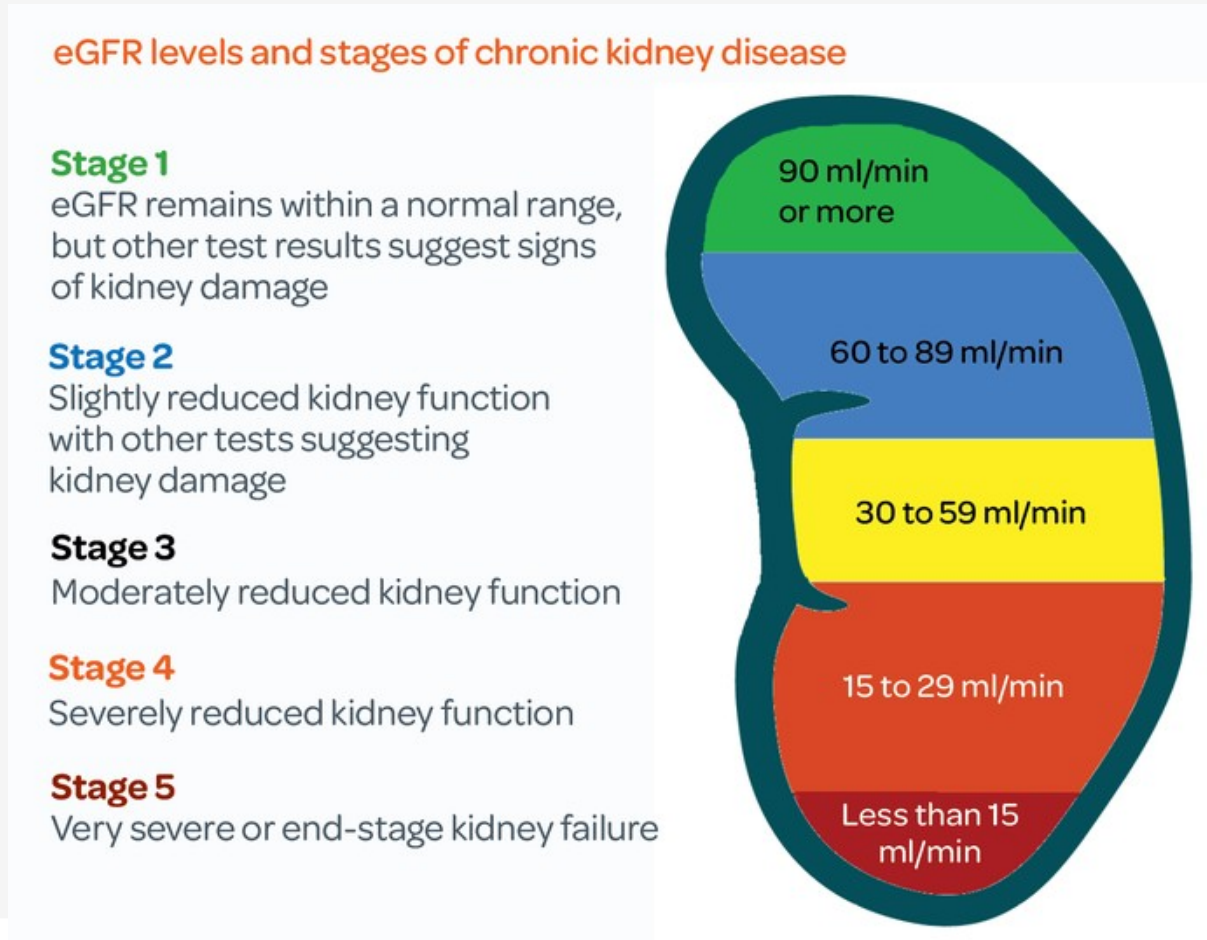


Fig : Stages of CKD

Urine Test

- ❖ Urine testing is a diagnostic tool used to assess various health including the kidneys conditions by analyzing the composition of urine.
- ❖ It helps in the detection of urinary tract infections, kidney function abnormalities, metabolic disorders, and other medical conditions.
- ❖ Some off the common test that are used in renal function test are :
Urinalysis, 24-Hour Urine Collection , Urine Creatinine Clearance, Urine Protein

Urinalysis

- ❖ Urinalysis is a diagnostic test that analyzes the composition of urine to assess various health conditions.
- ❖ Urine is responsible for removing waste, toxins, and excess fluids from the body, so its composition can reveal important insights into an individual's overall health.
- ❖ This test is commonly used to detect a range of diseases, including metabolic disorders, kidney dysfunction, and urinary tract infections.

Cont'd

❖ Urinalysis consists of the following components:

- **Macroscopic Examination** : volume, odor, color, specific gravity
- **Chemical Analysis** : protein content, glucose content, ketone content
- **Microscopic Examination** : Red blood cells, White blood cells, Epithelial cells, Crystals, Bacteria



Fig : Components of urinalysis

Urine protein

- ❖ Protein in urine test is often part of a urinalysis,
- ❖ Protein in urine may be an early sign of kidney disease before you have symptoms.
- ❖ A normal amount of protein in urine is less than 150 milligrams/day. However, the normal range may vary between laboratories.

Cont'd

- ❖ A protein level exceeding 150mg per day in urine is known as **proteinuria** which can be a sign of underlying health issues.
- ❖ Elevated protein in the urine may indicate serious conditions, including:
 - Chronic Kidney Disease (CKD)
 - Cardiovascular Risk
 - Autoimmune Disorders(lupus or Goodpasture's syndrome)
 - Kidney Cancer
 - Congestive Heart Failure

Clearance test

- ❖ Clearance refers to the volume of plasma completely cleared of a specific substance through excretion in urine per minute.
- ❖ It is a key measure particularly in determining how efficiently the kidneys are filtering waste products from the blood.
- ❖ The clearance of a substance can be calculated using the following formula:

$(U \times V)/P$ **where** U = concentration of substance in urine (mg/dl)

V = volume of urine excreted per minute (ml/min)

P = concentration of substance in plasma/serum (mg/dl)

Cont'd

- ❖ The types of clearance test can be generally categorized as follows

Types of Clearance tests	
Endogenous	Exogenous
Creatinine	Inulin
Cystatin C	Para-amino hippuric acid (PAHA)
	Diodrast (di-iodo pyridone acetic acid)

- ❖ Renal Clearance of Inulin is the **Gold Standard for determination of GFR.**
- ❖ It is fully filtered by the glomerulus and remains unaffected by both endogenous and exogenous factors throughout the nephron.

Cont'd

- ❖ **Cystatin C** is a low-molecular-weight, non-glycosylated protein produced consistently by all nucleated cells in the body.
- ❖ It is completely reabsorbed by the proximal tubules, with little to no cystatin excreted in urine.
- ❖ Normal circulating levels of cystatin C in adults is approximately **1 mg/l**.
- ❖ Cystatin C is considered a more accurate marker of kidney function than creatinine, particularly in the early stages of GFR decline.

Creatinine clearance test(CrCl)

- ❖ Creatinine clearance is a test used to estimate the kidney's ability to filter creatinine, a waste product, from the blood.
- ❖ Creatinine is cleared **almost exclusively through the glomerulus** thus used as a clinically useful measure of a patient's GFR.
- ❖ The test compares the amount of creatinine in the blood (serum) with the amount excreted in urine over a specified period

Cont'd

❖ The normal range is

Group	Normal Creatinine clearance
Male	97 – 137 ml/min
Female	88 – 128 ml/min

❖ Creatinine clearance may be used to estimate GFR for the following reason

- **Inulin** is not produced endogenously.(requires IV infusion)

Creatinine is an endogenous compound that fulfills these criteria.(as long as the muscle mass remains constant)

Factors influencing CrCl

1 Increased urine creatinine

- ❖ Acromegaly or Gigantism
- ❖ Diabetes mellitus
- ❖ Hypothyroidism
- ❖ Pregnancy
- ❖ Exercise

2 Decreased urine creatinine

- ❖ Hyperthyroidism
- ❖ Anemia
- ❖ Muscular dystrophy
- ❖ Advanced renal disease
- ❖ Leukemia

Case

- ❖ Mr. Bekele is a 62-year-old male with a history of hypertension (high blood pressure) for the past 10 years. He takes medication to manage his blood pressure, but he admits to not always taking it consistently. He has been experiencing increasing fatigue, swelling in his feet and ankles, and some back pain for the past few months. He reports increased nighttime urination. He denies any history of diabetes or kidney disease, although his father had kidney issues

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Physical Examination:

- Blood Pressure: 158/96 mmHg
- Mild swelling (edema) in lower legs and ankles.
- Otherwise, his physical exam is normal.

❖ You are asked

1. Interpret the Lab Results.
2. Potential Diagnosis.

Laboratory Results:

❖ Blood Tests:

- Serum Creatinine: 2.1 mg/dL (186 μ mol/L)
- BUN: 35 mg/dL (12.5 mmol/L)
- eGFR: 42 mL/min/1.73 m²
- Serum glucose: normal

❖ Urine Tests:

- Urinalysis: 2+ protein, no glucose, few red blood cells

Lab Interpretation

- ❖ **Serum Creatinine:** 2.1 mg/dL (186 μ mol/L) - This is elevated. (0.7-1.3 mg/dL for males)
- ❖ **BUN:** 35 mg/dL (12.5 mmol/L) - This is also elevated. (6-24mg/dL.)
- ❖ **eGFR:** 42 mL/min/1.73 m² - This is reduced and falls into CKD stage G3 (Moderate). (90 or above)
- ❖ **Urinalysis:** 2+ protein, few RBCs - The presence of 2+ protein (proteinuria) is abnormal. The few red blood cells (RBCs) are not highly significant and can be seen with many different conditions including mild infection or normal variance.

~~Potential Diagnosis~~

❖ Based on the combination of the history, physical findings, and lab results, the most likely potential diagnoses for Mr. Thomas are:

- **Chronic Kidney Disease (CKD):** This is the most probable diagnosis, given the **elevated serum creatinine and BUN, reduced eGFR and proteinuria**
- The fact that his father had kidney issues and that Mr Thomas has a history of hypertension increases the risk of CKD.
- The fatigue, swelling, and nighttime urination are also consistent with CKD.

Reference

- ❖ Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Workgroup. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl.* 2012;2:1–138.
- ❖ Spenser, H. (2017). Renal function tests and their interpretation. *Journal of Clinical Pathology*, 70(6), 456-461.

The background is a solid medium blue. It is decorated with various white abstract elements: wavy lines, small triangles, circles, and clusters of dots. The text 'THANK YOU' is centered in a white, hand-drawn, slightly irregular font.

THANK YOU

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